

MMM-NewLegoSets

THE COMPLETE 2.0 USER GUIDE

New LEGO sets. Your layout. Your colors. Your motion.

10 layouts | 11 themes | 22 animations | 1-10 visible products



SEPARATE CLOCKS

Poll LEGO.com every few hours while slides move every few seconds.

HONEST DATES

Shipping estimates are never mislabeled as product release dates.

MIRROR READY

Responsive container layouts and reduced-motion support are built in.

CACHE RESCUE

A failed request keeps the last good product list visible and labeled.

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This manual accompanies the exhaustive README in the module folder. It focuses on setup, visual choices, common recipes, and quick reference.

1. Install and start

Place the folder at **MagicMirror/modules/MMM-NewLegoSets**. There are no npm runtime dependencies. Add a module entry to MagicMirror's config and restart MagicMirror.

```
{
  module: "MMM-NewLegoSets",
  position: "top_right",
  config: {
    productCount: 1,
    layout: "hero",
    theme: "lego"
  }
}
```

What the first run does

1. FETCH

The Node helper requests LEGO.com's new-products listing.

2. NORMALIZE

Product image, price, pieces, number, and availability are normalized.

3. CACHE

A successful pool is stored as last-known-good data.

4. CYCLE

The browser displays 1-10 products and advances independently.

The two clocks

Setting	Default	Meaning
data.pollInterval	6 hours	Download a fresh list from LEGO.com.
cycle.interval	12 seconds	Change the visible set/group without downloading.

Use milliseconds. For readability, write $4 * 60 * 60 * 1000$ for four hours and $15 * 1000$ for fifteen seconds.

2. Choose a layout

The arrangements cover familiar product-display patterns. Auto is the safest starting point; it selects hero, list, or grid based on product count.

AUTO Hero for 1, list for 2-3, grid for 4-10.	HERO One large image/details presentation.
LIST Stacked image-and-details rows.	GRID Uniform multi-column product cards.
COMPACT Dense thumbnail rows for small regions.	SPLIT Wide image-led cards.
CAROUSEL Main product plus preview rail.	FILMSTRIP One horizontal sequence.
MASONRY Flowing columns for varied metadata.	TABLE Aligned, dense, highly scannable rows.

Sizing controls

Setting	Use
productCount	Visible products, 1 through 10.
layoutSettings.columns	Grid and masonry columns, 1 through 10.
moduleWidth / moduleMaxWidth	Region width and maximum pixel width.
cardMinWidth	Minimum grid-card width before space becomes tight.
gap / padding / cardPadding	Density and breathing room.
imagePosition / alignItems	Card composition and vertical alignment.

3. Choose a theme

Every layout consumes the same theme variables, so switching a theme changes cards, text, accents, borders, background, and decorations together.

Theme	Palette	Decoration
mirror	Black / white	None
lego	Red / yellow / blue	Corner bricks
tuxedo	Black / silver / white	Bow tie
pride	Rainbow	Rainbow bars
progress	Progress Pride colors	Chevron
trans	Blue / pink / white	Flag bars
bisexual	Pink / purple / blue	Flag bars
lesbian	Orange / white / pink	Flag bars
nonbinary	Yellow / white / purple / black	Flag bars
pansexual	Pink / yellow / cyan	Flag bars
custom	Your values	Selectable



Set `showThemeDecorations: false` to keep a preset's palette and remove its corner artwork. Use theme: "custom" to control background, surface, text, muted, accent, accent2, border, shadow, gradient, and decoration.

4. Motion and LEGO build

Twenty-two choices cover restrained fades through expressive brick construction. The module honors the operating system's reduced-motion preference by default.

SUBTLE none, fade, crossfade, blur	DIRECTIONAL slideLeft, slideRight, slideUp, slideDown
DEPTH zoomIn, zoomOut, flipX, flipY	PLAYFUL rotate, roll, bounce, swing, elastic
REVEALS wipe, shutter	BRICKS legoBuild, legoBreakBuild, random

```
animation: {
  name: "legoBreakBuild",
  duration: 1100,
  easing: "cubic-bezier(0.22, 1, 0.36, 1)",
  stagger: 18,
  particleCount: 36,
  brickSize: 10,
  respectReducedMotion: true
}
```

Performance recipe

For Raspberry Pi-class mirrors, start with fade, 600-800 ms, and 12-24 particles. High particle counts, blur, and large shadows increase GPU work. Keep reduced-motion support on unless you know the viewer's needs.

5. Data, cycling, and caching

Data key	Default	Explanation
locale / countryCode	en-us / US	Storefront, formatting, and Brickset region.
pageCount	2	Listing pages attempted, 1-8.
poolSize	10	Products kept for cycling, maximum 50.
includeComingSoon	false	Include coming-soon status.
includePreorders	false	Include preorder status.
pollInterval	6 hours	Successful refresh clock.
retryInterval	10 minutes	Failure retry clock.
requestTimeout	20 seconds	Network timeout per request.
cacheEnabled	true	Persist the last good pool.
cacheMaxAge	7 days	Preferred freshness window.
sortBy / direction	source / desc	Source, dates, price, pieces, number, or name.

Cycle key	Default	Explanation
enabled	true	Automatic slide movement.
interval	12 seconds	Time between visible changes.
step	1	Positions advanced each cycle.
loop	true	Wrap at the end.
shuffle	false	Shuffle each fresh pool.
pauseOnHover	false	Pause for pointer/touchscreen inspection.
indicatorStyle	dots	Dots, bars, numbers, or none.

Single-product slideshow

```
productCount: 1,  
data: { poolSize: 10, pollInterval: 6 * 60 * 60 * 1000 },  
cycle: { enabled: true, interval: 12 * 1000, step: 1 }
```

6. Configure every field

Available fields: image, name, price, pieceCount, setNumber, releaseDate, announcedDate, availability, ageRange, and pricePerPiece.

Image controls

Key	Meaning
show	Show or hide the product image.
width / height	Preferred box size, default 180 x 180 px.
minWidth / maxWidth	Responsive limits.
aspectRatio	CSS aspect ratio, default 1 / 1.
fit / position	Contain, cover, fill, scale-down and alignment.
opacity	0 is transparent; 1 is opaque.
background / borderRadius	Image stage appearance.
filter	Any valid CSS filter.
showPlaceholder	LEGO-colored fallback if an image is absent.

Text controls

Keys	Meaning
show, order	Visibility and ordering.
label, prefix, suffix	Value wording.
missingText	Fallback; empty hides unavailable data.
fontFamily, fontSize, fontWeight, fontStyle	Field-specific typography.
letterSpacing, lineHeight, lineClamp	Text density and truncation.
textTransform, textAlign	Case and alignment.
color, opacity	Field color/transparency.
effect	none, shadow, outline, glow, neon, gradient, or letterpress.

7. Dates without guesswork

The module deliberately preserves unknowns. This prevents plausible-looking but false product history.

LEGO LISTING

Explicit available/release/launch dates may be used.

NOT A RELEASE DATE

Will ship by and backorder estimates are ignored as release dates.

BRICKSET

Optional regional dateFirstAvailable or launchDate enrichment.

ANNOUNCEMENTS

Supply manually; LEGO's listing has no dependable announcement field.

```
data: {
  bricksetApiKey: "YOUR_KEY",
  countryCode: "US",
  metadataOverrides: {
    "11512": {
      releaseDate: "2026-08-01",
      announcedDate: "2026-06-10"
    }
  }
}
```

Cache discovery time is stored internally but is never displayed as an announcement date. Missing date fields stay hidden unless their field's missingText is configured.

8. Ready-made recipes

Ten-product Progress Pride grid

```
layout: "grid",
productCount: 10,
theme: "progress",
data: { poolSize: 20 },
cycle: { interval: 30000, step: 10, indicatorStyle: "bars" },
layoutSettings: { columns: 2, moduleMaxWidth: 900 }
```

Tuxedo information table

```
layout: "table",
productCount: 8,
theme: "tuxedo",
animation: { name: "wipe", duration: 700 },
typography: { baseFontSize: 14 },
fields: {
  releaseDate: { show: false },
  announcedDate: { show: false },
  availability: { show: false }
}
```

Transparent images, no motion

```
layout: "list",
productCount: 4,
theme: "mirror",
cycle: { enabled: false },
animation: { name: "none" },
fields: { image: { opacity: 0.55, filter: "saturate(.8)" } }
```

9. Validation and troubleshooting

Run the offline validator after every config edit. Run live validation after updates or when LEGO.com output looks wrong.

```
cd ~/MagicMirror/modules/MMM-NewLegoSets
npm run validate
npm run validate:live
```

Symptom	Check
Only one set	Raise productCount to show more; or keep 1 and ensure poolSize > 1 for cycling.
Sets never change	cycle.enabled, cycle.interval, and poolSize larger than productCount.
Images blank	Access to www.lego.com/cdn and DNS filtering.
Dates blank	Expected without trustworthy data; use Brickset or overrides.
Motion missing	The operating system may request reduced motion.
Too many dots	Raise cycle.step, switch to bars, or hide indicators.
Narrow overflow	Use auto/compact/table, fewer columns, or smaller cardMinWidth.
Custom font missing	Use a reachable WOFF2 path, preferably local.
Brickset warning	Check key/connectivity; products still render without enrichment.
Listing parser error	Run validate:live and inspect MagicMirror's terminal.

Validated release matrix

10 desktop layouts + 10 narrow layouts; all 11 themes; product counts 1-10; all 22 animation states; 36-particle LEGO build; real single-product advancement; current and legacy LEGO payloads; live 10-product fetch; syntax and boundary validation.

10. Quick reference

Top-level defaults

Key	Default
title / subtitle	New LEGO Sets / Recently released
layout / productCount	auto / 1
theme / decorations	lego / true
dateFormat	year numeric, month short, day numeric
debug	false

Layout defaults

Key	Default
columns / gap	2 / 12
cardMinWidth	170
moduleWidth / maxWidth	auto / 760
padding / cardPadding	12 / 10
borderRadius / cardBorderRadius	8 / 6
imagePosition / alignItems	left / center
header / footer / counter / updated	all true
source	false

Typography defaults

Key	Default
fontFamily	Roboto, Arial, sans-serif
customFontName / Url	empty
baseFontSize / align	16 / left
textEffect / antialias	none / true

Keep building

The module is designed to be changed. Start with product count, layout, and theme. Then tune cycle timing and animation. Finally, adjust fields and typography once the information density feels right on the physical mirror.

START SIMPLE

auto + one product + your preferred theme

WATCH THE CLOCKS

pollInterval fetches; cycle.interval slides

PRESERVE UNKNOWNNS

Blank dates are better than invented history

VALIDATE

Run both validators before relying on an upgrade

Documentation and community references

MagicMirror core module, rendering, and animation documentation; Brickset API v3 documentation; MMM-Carousel; and MagicMirror community discussion of CSS animation performance. Full links are in README.md.

MMM-NewLegoSets is unofficial and is not affiliated with or endorsed by the LEGO Group or Brickset. LEGO and product names are trademarks of the LEGO Group. LGBTQ+ theme names identify color presets and do not claim ownership of flags or communities.